

Ariyan Zarei

Machine Learning Research Scientist at Meta

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Experience

Research Scientist

Meta Reality Labs

May 2023 – Ongoing

Burlingame, CA

- Led the development of a large-scale human full-body motion tracking and geometry registration pipeline for the Codec Avatar program, used to process billion-scale internet in-the-wild data.
- Designed and developed human hand tracking pipelines and processed multi-view studio and monocular mobile datasets.

Machine Learning Software Engineer Intern

Meta Inc.

May 2022 – August 2022

Menlo Park, CA

- Worked with the Ads Fairness team on modeling fairness strategies.
- Designed and developed a policy-gradient reinforcement learning controller and an aggregated supervised learning model to address potential miscalibrations between different groups of users.

Graduate Research Assistant

University of Arizona

January 2019 – April 2023

Tucson, AZ

- Developed an efficient and scalable geo-correction and image stitching method, MegaStitch, for orthomosaicing agricultural images in order to estimate relations between plant genotypes and phenotypes. Joint work with the PhytoOracle team and CyVerse.
- Developed MegaStitch3D for geo-correcting 3D point clouds. Joint work with the PhytoOracle team and CyVerse.
- Developed PlantSegNet, an end-to-end deep instance segmentation model for segmenting 3D point clouds of sorghum leaves. Joint work with the PhytoOracle team and CyVerse.
- Designed and developed NowCasting-Nets, neural networks for estimating precipitation patterns on spatio-temporal data using the IMERG product.

IT Intern

NP Photonics

June 2021 – March 2022

Tucson, AZ

- Developed multiple enterprise applications for data management, ticketing, LMS, and internal workflows using Python/Qt, Microsoft SQL Server, and Office 365, including Power Apps, SharePoint, and Power Automate.
- Managed and maintained local network infrastructure with Windows Server and Active Directory.

Graduate Teaching Assistant

University of Arizona

August 2018 – May 2019

Tucson, AZ

Undergraduate Research Assistant

Shahid Beheshti University

January 2016 – December 2017

Tehran, Iran

- Developed gesture recognition software using Microsoft Kinect to assist in treating patients with neurodevelopmental disorders, such as autism and hyperactivity.

Education

Ph.D. in Computer Science

University of Arizona

Advisor: Prof. Kobus Barnard

August 2018 – May 2023

GPA: 3.80

M.Sc. in Computer Science

University of Arizona

August 2018 – December 2020

GPA: 3.80

B.S. in Computer Science

Shahid Beheshti University

August 2013 – April 2018

GPA: 3.92

Publications

1. Ke Li, Robin Kips, Filippo Arcadu, Marek Schikora, John Ho, Chris Graf, Jim Chen, Xingxing Cai, Gawsalyan Sivapalan, Ariyan Zarei, et al. “Head and Body Pose Estimation using Ultrasound Sensing on Smart Glasses.” *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*, pp. 1–6, 2026.
2. Junxuan Li, Rawal Khirodkar, Chengan He, Zhongshi Jiang, Giljoo Nam, Lingchen Yang, Jihyun Lee, Egor Zakharov, Zhaoen Su, Rinat Abdrashitov, et al. “Large-scale Codec Avatars: The Unreasonable Effectiveness of Large-scale Avatar Pretraining.” *arXiv preprint arXiv:2604.02320*, 2026.
3. Ariyan Zarei, Emmanuel Gonzalez, Nirav Merchant, Duke Pauli, Eric Lyons, and Kobus Barnard. “MegaStitch: Robust Large-Scale Image Stitching.” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–9, 2022.
4. Mohammad Reza Ehsani, Ariyan Zarei, Hoshin Vijai Gupta, Kobus Barnard, Eric Lyons, and Ali Behrangi. “NowCasting-Nets: Representation Learning to Mitigate Latency Gap of Satellite Precipitation Products Using Convolutional and Recurrent Neural Networks.” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–21, 2022.
5. Ariyan Zarei, Bosheng Li, James C. Schnable, Eric Lyons, Duke Pauli, Kobus Barnard, and Bedrich Benes. “PlantSegNet: 3D point cloud instance segmentation of nearby plant organs with identical semantics.” *Computers and Electronics in Agriculture*, vol. 221, p. 108922, 2024.
6. Emmanuel Gonzalez, Ariyan Zarei, Nathaniel Hendler, Michele Cosi, Jeffrey Demieville, Sebastian Calleja, Travis Simmons, Holly Ellingson, Nirav Merchant, Eric Lyons, et al. “PhytoOracle: Scalable, modular phenomic data processing pipelines.” 2021.
7. Emmanuel M. Gonzalez, Ariyan Zarei, Sebastian Calleja, Clay Christenson, Bruno Rozzi, Jeffrey Demieville, Jiahuai Hu, Andrea L. Eveland, Brian Dilkes, Kobus Barnard, et al. “Quantifying leaf symptoms of sorghum charcoal rot in images of field-grown plants using deep neural networks.” *The Plant Phenome Journal*, vol. 7, no. 1, e20110, 2024.
8. Wenting Luo, Emmanuel Gonzalez, Ariyan Zarei, Sebastian Calleja, Bruno Rozzi, Jeffrey Demieville, Haiquan Li, Maria-Jose Truco, Dean Lavelle, Richard Micheltore, et al. “Leaf cuticular wax composition of a genetically diverse collection of lettuce (*Lactuca sativa* L.) cultivars evaluated under field conditions.” *Heliyon*, vol. 10, no. 5, 2024.
9. Nathaniel Hendler, Emmanuel Gonzalez, Ariyan Zarei, Travis Simmons, Jeffrey Demieville, Sebastian Calleja, Eric Lyons, and Duke Pauli. “Improvements on Multiway ICP Registration for Reconstructing Individual Plants from 3D Field Scans.” *Authorea Preprints*, 2022.

Awards and Leadership

- Led two student clubs during studies at the University of Arizona, ICAT, and Shahid Beheshti University, SACSS.
- Received the Tech Launch Arizona Award to design and develop an intelligent gardening application.
- Achieved a bronze medal in the WorldSkills Competition in IT Software Solutions for Business.
- Received the 2022 Galileo Circle Scholarship from the School of Science, University of Arizona.

Expertise

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| • Digital Humans | • 3D Computer Vision |
| • Human Motion Tracking and Estimation | • Neural Rendering |
| • Geometry Processing | • Photogrammetry |

Technical Skills

Machine Learning and Computer Vision	PyTorch, PyTorch3D, Keras, TensorFlow, OpenCV
AI-Assisted Development	Claude, Claude CLI, Codex; agentic coding workflows for building large-scale projects with minimal manual coding
Software	Python, Django REST Framework, React Native, Docker, Singularity, Java, C#, C, C++, Microsoft SQL Server
Hardware Prototyping	MicroPython, Raspberry Pi Pico, Raspberry Pi, ESP32, embedded circuits, personal hardware projects
Systems	Windows Server, Active Directory, Office 365, Power Apps, SharePoint, Power Automate

Languages

English; Persian

Referees

Prof. Kobus Barnard	University of Arizona, kobus.ca
Dr. Eric Lyons	University of Arizona, cals.arizona.edu/spls/content/eric
Dr. Duke Pauli	University of Arizona, cals.arizona.edu/spls/content/duke
Prof. Bedrich Benes	Purdue University, cs.purdue.edu/homes/bbenes/cv
Dr. Ali Katanforoush	Shahid Beheshti University, facultymembers.sbu.ac.ir/katanforoush